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(54) ELECTRIC COMPRESSOR

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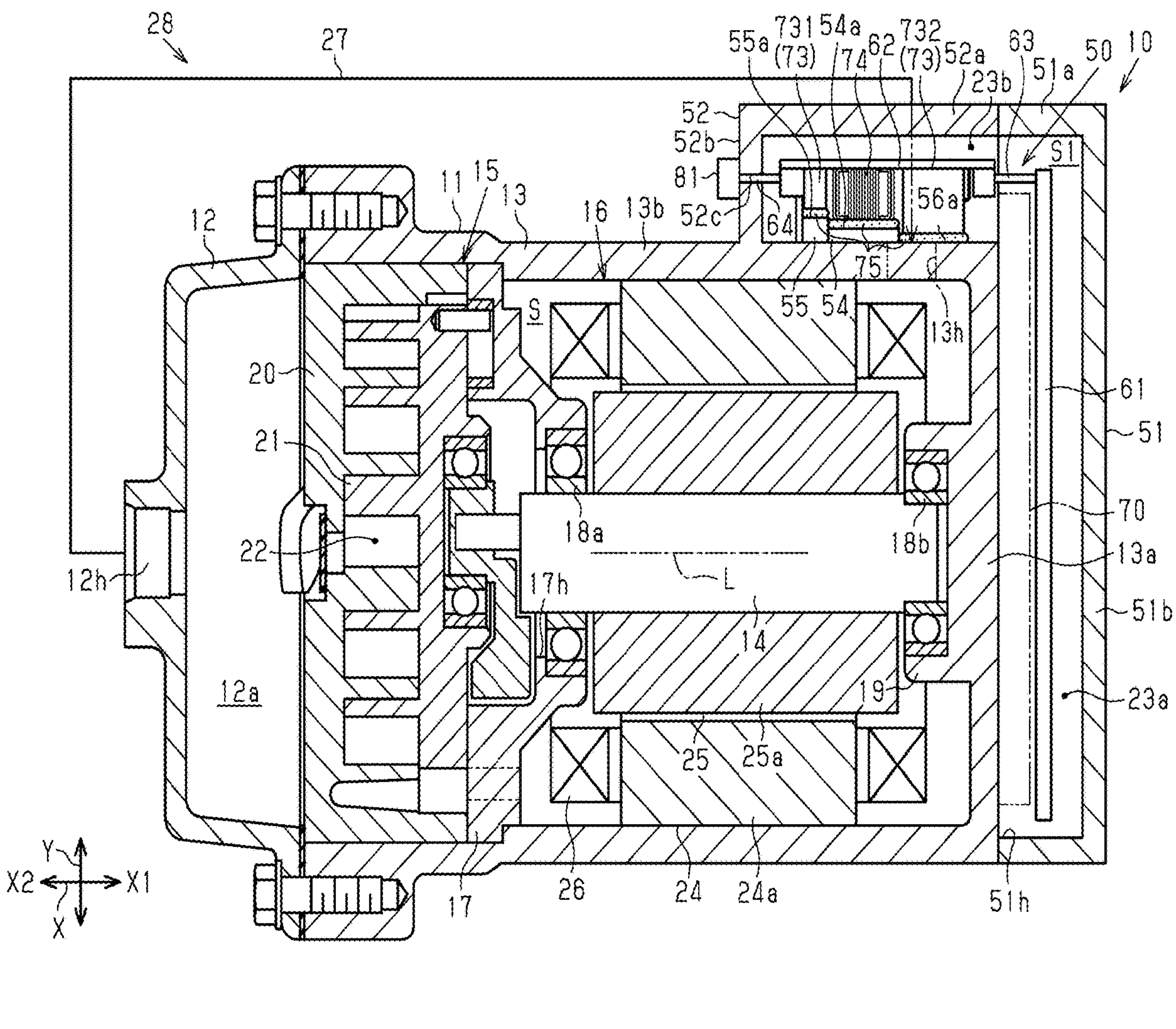
(57) ABSTRACT

An electric compressor includes: a rotary shaft; an inverter that includes a second circuit board having a plurality of electronic components; and a housing that includes a motor housing. The motor housing has a projected portion projecting outward from the motor housing in a radial direction of the rotary shaft, having a bottomed-cylindrical shape with an opening, and extending in an insertion direction of the second circuit board. The electronic components protrude from the second circuit board and are arranged in the insertion direction in ascending order of protrusion height from the second circuit board. The projected portion has a protrusion portion facing the electronic components and protruding toward the electronic components in the radial direction, with protrusion height of the protrusion portion increasing in the insertion direction correspondingly with an arrangement of the electronic components. A heat transfer member is provided between the protrusion portion and the electronic components.



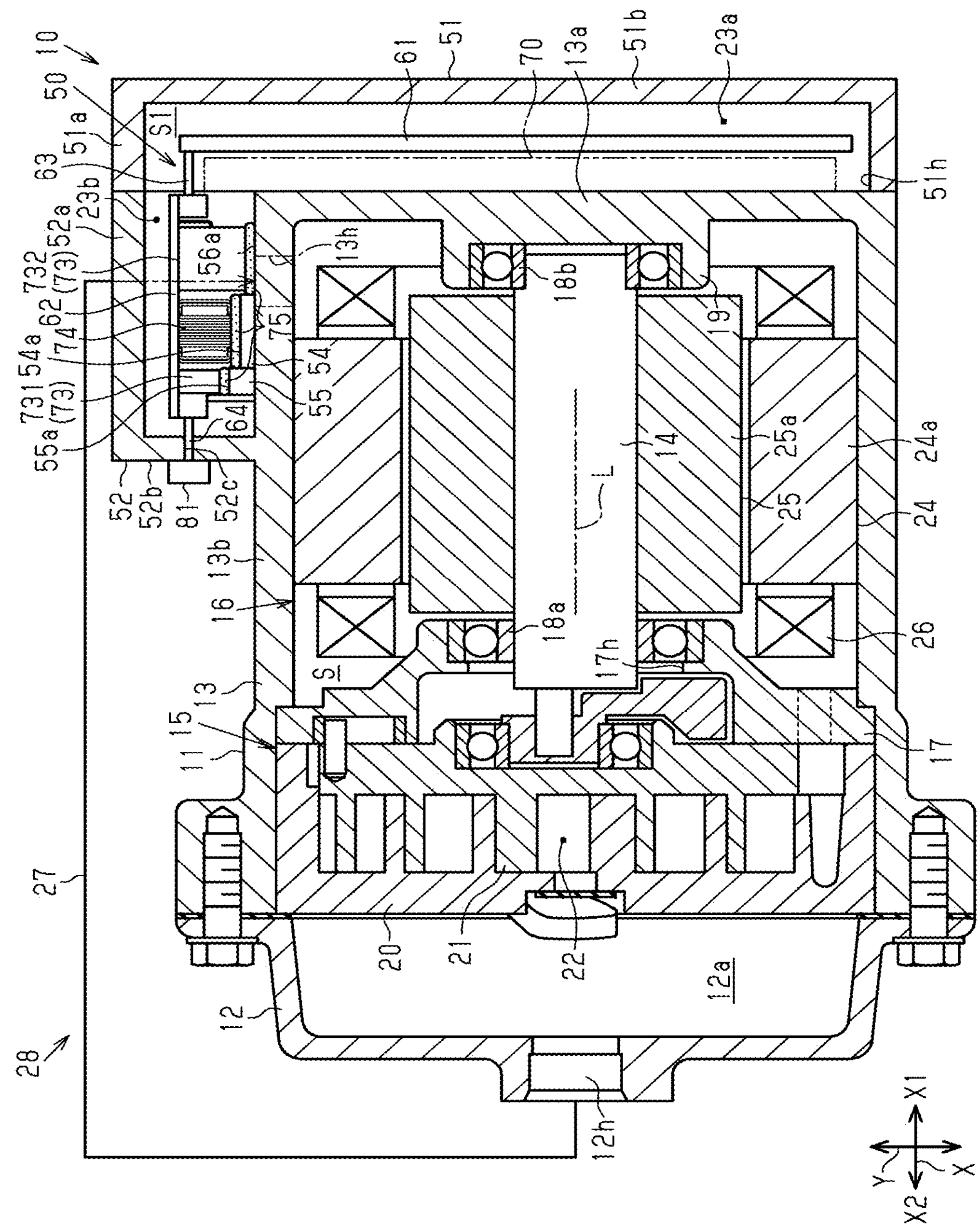
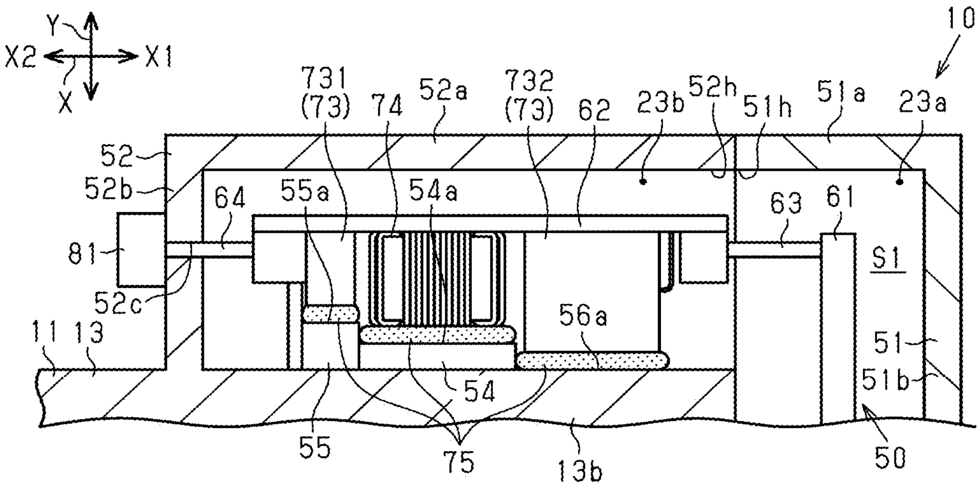

10
50
51a
63
23b
52a
732
(73)
62
74
54a
731
(73)
55a
52
52b
81
52c
64
55
54
75
56a
13h
18b
19
S1
61
51
70
13a
51b
23a
51h
14
L
25a
24a
25
24
13b
16
18a
17h
26
S
13
15
11
17
27
20
21
22
12a
12
28
12h
X1
X2
X
Y

FIG. 1

FIG. 2



Y
X2
X1
X
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(73)
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(73)
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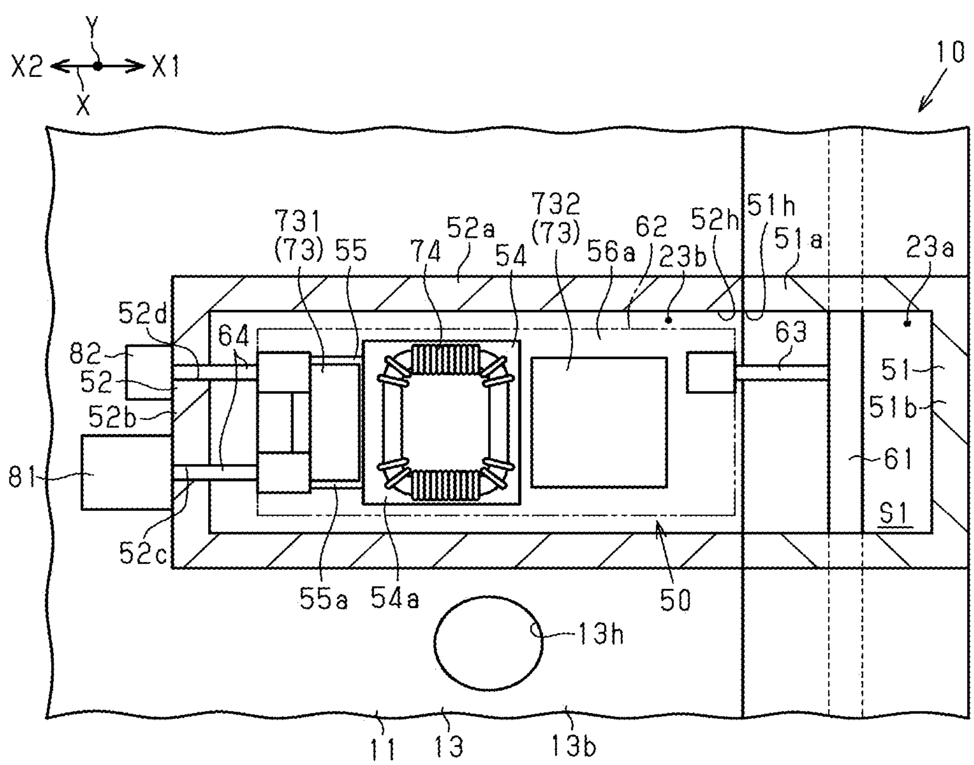
23b
52h
51h
51a
23a
52
52b
64
55a
54a
63
61
81
S1
52c
56a
11
13
51
51b
54
55
75
13b
50

FIG. 3



Y
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23b
52h
51h
51a
23a
52d
64
63
82
52
51
52b
51b
61
81
S1
52c
55a
54a
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13h
11
13
13b

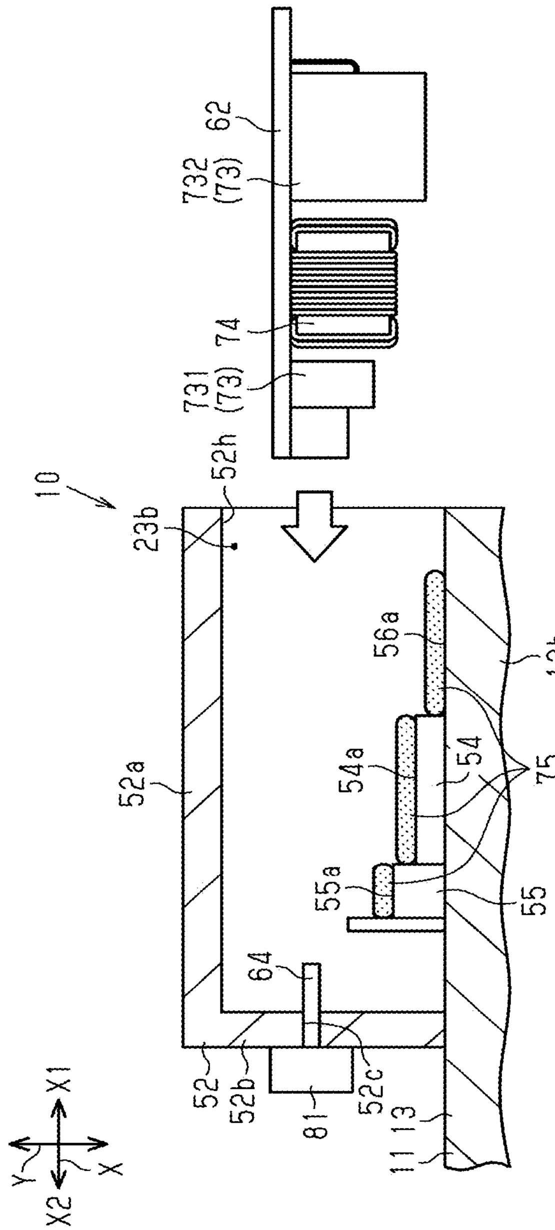
62
732
(73)
74
731
(73)
10
52h
23b
56a
13b
54a
54
75
52a
55a
55
64
52
52b
52c
81
11
13
X1
X2
X
Y

FIG. 4

ELECTRIC COMPRESSOR

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-028332 filed on Feb. 28, 2024, the entire disclosure of which is incorporated herein by reference.

[0002] The present disclosure relates to an electric compressor.

BACKGROUND ART

[0003] Japanese Patent Application Publication No. 2014-058910 discloses an electric compressor that includes a rotary shaft, a compression part configured to compress a fluid with rotation of the rotary shaft, a motor configured to drive the compression part, an inverter configured to drive the motor, and a housing. The housing accommodates the rotary shaft, the compression part, the motor, and the inverter. The inverter includes a first circuit board and a second circuit board. The first circuit board has, on the first circuit board, a driving element for driving the motor. The second circuit board has, on the second circuit board, an electronic component for removing noise from the first circuit board. The first circuit board is accommodated in a first accommodation chamber, and the second circuit board is accommodated in a second accommodation chamber.

[0004] Such an electric compressor may include a heat transfer member disposed between the electronic component of the second circuit board and a fixing surface of the second accommodation chamber to which the electronic component is fixed. When the second circuit board and the electronic component are mounted to the housing, the second circuit board and the electronic component may be moved in the axial direction of the rotary shaft in the second accommodation chamber. If the electronic component is moved while coming into contact with the fixing surface of the second accommodation chamber, the thickness of the heat transfer member may exhibit unevenness.

[0005] The present disclosure is directed to providing an electric compressor in which thickness unevenness of a heat transfer member is reduced.

SUMMARY

[0006] In accordance with an aspect of the present disclosure, there is provided an electric compressor that includes: a rotary shaft; a compression part; a motor configured to drive the compression part; an inverter; and a housing for accommodating the rotary shaft, the compression part, the motor, and the inverter. The compression part is configured to compress a fluid with rotation of the rotary shaft. The inverter is configured to drive the motor. The inverter includes a first circuit board and a second circuit board. The first circuit board has, on the first circuit board, a driving element for driving the motor. The second circuit board has, on the second circuit board, an electronic component for removing noise from the first circuit board. The housing includes: a motor housing having a suction chamber in which the motor is accommodated and into which the fluid is drawn; a compression part housing in which the compression part is accommodated and from which the fluid compressed is discharged; an inverter housing having an inverter accommodation chamber in which the inverter is accommodated; and a partition wall that separates the suction chamber from the inverter accommodation chamber. The motor housing has a projected portion projecting outward from the motor housing in a radial direction of the rotary shaft and having a bottomed-cylindrical shape with an opening that opens in an axial direction of the rotary shaft. The projected portion extends in an insertion direction of the second circuit board in which the second circuit board is inserted into the projected portion through the opening from a side where the partition wall is located. The electronic component includes a plurality of the electronic components that protrude from the second circuit board and are arranged in the insertion direction of the second circuit board toward a bottom portion of the projected portion in ascending order of protrusion height from the second circuit board. The projected portion has a protrusion portion facing the electronic components and protruding toward the electronic components in the radial direction of the rotary shaft, with protrusion height of the protrusion portion increasing in the insertion direction of the second circuit board toward the bottom portion of the projected portion correspondingly with an arrangement of the electronic components. A heat transfer member is provided between the protrusion portion and the electronic components.

[0007] Other aspects and advantages of the disclosure will become apparent from the following description, taken in conjunction with the accompanying drawings, illustrating by way of example the principles of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The disclosure, together with objects and advantages thereof, may best be understood by reference to the following description of the embodiments together with the accompanying drawings in which:

[0009] FIG. **1** is a sectional view of an electric compressor according to an embodiment;

[0010] FIG. **2** is a sectional view of a part of the electric compressor;

[0011] FIG. **3** is a sectional view of a part of the electric compressor; and

[0012] FIG. **4** is a sectional view of a part of the electric compressor, explaining a method of mounting a second circuit board, a capacitor, and a magnetic component to a housing.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0013] The following will describe an embodiment of an electric compressor with reference to accompanying FIGS. **1** to **4**. The electric compressor of the embodiment is applied, for example, to a vehicle air conditioner.

[0014] FIG. **1** illustrates an electric compressor **10** that includes a housing **11**. The housing **11** includes a discharge housing **12**, a motor housing **13**, and an inverter housing **51**. The discharge housing **12**, the motor housing **13**, and the inverter housing **51** are each made of a metal material. The discharge housing **12**, the motor housing **13**, and the inverter housing **51** are each made of aluminum, for example. The discharge housing **12** has a cylindrical shape. The motor housing **13** includes an end wall **13*a*** having a plate-like shape, a peripheral wall **13*b*** having a cylindrical shape, and a projected portion **52**. The peripheral wall **13*b*** has a cylindrical shape and extends from an outer peripheral portion of the end wall **13***a*. The projected portion **52** has a bottomed-cylindrical shape.

[0015] The electric compressor **10** includes a rotary shaft **14**. The rotary shaft **14** is accommodated in the motor housing **13**.

[0016] The electric compressor **10** includes a compression part **15**, a motor **16**, and an inverter **50**. The compression part **15** and the motor **16** are accommodated in the motor housing **13**. The compression part **15** is driven by rotation of the rotary shaft **14**. The compression part **15** compresses a refrigerant as a fluid with the rotation of the rotary shaft **14**. The motor **16** rotates the rotary shaft **14** to drive the compression part **15**. The compression part **15** and the motor **16** are arranged side by side in an axial direction X of the rotary shaft **14**, which is indicated by an axis L. The motor **16** is disposed between the compression part **15** and the end wall **13***a* of the motor housing **13**.

[0017] The electric compressor **10** includes a shaft support member **17**. The shaft support member **17** is disposed between the compression part **15** and the motor in the motor housing **13**. The shaft support member **17** has an insertion hole **17***h*. The insertion hole **17***h* is formed at a center portion of the shaft support member **17**. A first end of the rotary shaft **14** is inserted through the insertion hole **17***h*. A bearing **18***a* is disposed between the insertion hole **17***h* and the first end of the rotary shaft **14**. The first end of the rotary shaft **14** is rotatably supported by the shaft support member **17** via the bearing **18***a*.

[0018] The motor housing **13** has a bearing holding portion **19** having a cylindrical shape. The bearing holding portion **19** protrudes from a center portion of the end wall **13***a* of the motor housing **13**. A second end of the rotary shaft **14** is inserted into the bearing holding portion **19**. A bearing **18***b* is disposed between the bearing holding portion **19** and the second end of the rotary shaft **14**. The second end of the rotary shaft **14** is rotatably supported by the bearing holding portion **19** via the bearing **18***b*.

[0019] The compression part **15** includes a fixed scroll **20** and a movable scroll **21**. The fixed scroll **20** is fixed to an inner peripheral surface of the peripheral wall **13***b* of the motor housing **13**. The movable scroll **21** faces the fixed scroll **20**. The fixed scroll **20** and the movable scroll **21** mesh with each other. The fixed scroll **20** and the movable scroll **21** cooperate to define a compression chamber **22** with variable volume. The compression chamber **22** compresses the refrigerant drawn, and discharges the refrigerant.

[0020] The motor **16** includes a stator **24** having a cylindrical shape and a rotor **25** having a cylindrical shape. The rotor **25** is disposed inside the stator **24**. The rotor **25** rotates together with the rotary shaft **14**. The stator **24** surrounds the rotor **25**. The rotor **25** includes a rotor core **25***a* fixed to the rotary shaft **14**, and a plurality of permanent magnets (not illustrated) disposed in the rotor core **25***a*. The stator **24** includes a cylindrical stator core **24***a* and a coil **26** wound around the stator core **24***a*. When power is supplied to the coil **26**, the rotor **25** and the rotary shaft **14** rotate.

[0021] The motor housing **13** has a suction port **13***h*. The suction port **13***h* is formed in the peripheral wall **13***b*. The refrigerant is drawn into the motor housing through the suction port **13***h*. The suction port **13***h* is connected to a first end of an external refrigerant circuit **27**. The discharge housing **12** has a discharge chamber **12***a*. The discharge housing **12** has a discharge port **12***h*. The discharge port **12***h* is communicated with the discharge chamber **12***a*. The discharge port **12***h* is connected to a second end of the external refrigerant circuit **27**.

[0022] The refrigerant is drawn from the external refrigerant circuit **27** into the motor housing **13** through the suction port **13***h*. The motor housing **13** has a suction chamber S. In other words, the motor housing **13** has the suction chamber S in which the motor **16** is accommodated and into which the fluid is drawn. The refrigerant drawn into the suction chamber S in the motor housing **13** is further ( ) drawn into the compression chamber **22** through the orbiting motion of the movable scroll **21**. The refrigerant in the compression chamber **22** is compressed with the orbiting motion of the movable scroll **21**. The refrigerant compressed in the compression chamber **22** is then discharged into the discharge chamber **12***a*. That is, the refrigerant compressed by the compression part **15** is discharged from the motor housing **13** to the discharge chamber **12***a*. Accordingly, the motor housing also serves as the compression part housing of the present disclosure from which the fluid compressed is discharged. The discharge chamber **12***a* is a discharge pressure region. The refrigerant discharged to the discharge chamber **12***a* flows out to the external refrigerant circuit **27** through the discharge port **12***h*. The refrigerant flows through a heat exchanger and an expansion valve of the external refrigerant circuit **27**, and returns to the motor housing **13** through the suction port **13***h*. The electric compressor **10** and the external refrigerant circuit **27** cooperate to form a vehicle air conditioner **28**.

[0023] The inverter housing **51** has an open end that is open toward the end wall **13***a* and the projected portion **52**. The open end of the inverter housing **51** is attached to the end wall **13***a* and the projected portion **52**, so that the inverter housing **51** is attached to the motor housing **13**. The inverter housing **51** and the projected portion **52** cooperate to define a space that partly serves as an inverter accommodation chamber S1 for accommodating the inverter **50**. That is, the housing **11** includes the inverter housing **51** that defines the inverter accommodation chamber S1 for accommodating the inverter **50**. The end wall **13***a* of the motor housing **13** serves as the partition wall of the present disclosure that separates the suction chamber S from the inverter accommodation chamber S1.

[0024] The axial direction X includes opposite directions that serve as a first axial direction **X1** and a second axial direction **X2**, respectively. In the axial direction X, the first axial direction **X1** is oriented toward a first side, and the second axial direction **X2** is oriented toward a second side. The inverter housing **51** has a first cylindrical portion **51***a* having a cylindrical shape and extending in the axial direction X, and a closing portion **51***b*. The first cylindrical portion **51***a* has opposite ends **10**) that define the first cylindrical portion **51***a* in the axial direction X. One of the opposite ends of the first cylindrical portion **51***a* on the first side is closed by the closing portion **51***b*. The first cylindrical portion **51***a* has, for example, a cylindrical shape. The closing portion **51***b* has, for example, a flat plate shape and extends perpendicular to the axial direction X. The inverter housing **51** has a first open portion **51***h* that opens in the second axial direction **X2**. The first open portion **51***h* is an open end of the first cylindrical portion **51***a* that opens in the second axial direction **X2**.

[0025] As illustrated in FIGS. **2** and **3**, the projected portion **52** has a second cylindrical portion **52***a* having a cylindrical shape and extending in the axial direction X, and a bottom portion **52***b*. The second cylindrical portion **52***a* has opposite ends that define the second cylindrical portion **52***a* in the axial direction X. One of the opposite ends of the second cylindrical portion **52***a* on the second side is closed by the bottom portion **52***b*. The bottom portion **52***b* has, for example, a flat plate shape and extends perpendicular to the axial direction X. The bottom portion **52***b* has a first through hole **52***c* and a second through hole **52***d*. The projected portion **52** has a second open portion **52***h* that opens in the first axial direction X**1**. The second open portion **52***h* is an open end of the second cylindrical portion **52***a* on the first side. In other words, the second open portion **52***h* serves as the opening of the present disclosure through which the second circuit board **62** is inserted.

[0026] As illustrated in FIG. **1**, in this embodiment, a part of the second cylindrical portion **52***a* is connected to the peripheral wall **13***b* of the motor housing **13**. The second cylindrical portion **52***a* may be integrally formed with the peripheral wall **13***b*, or may be separately formed from and connected to the peripheral wall **13***b*. Furthermore, the part of the second cylindrical portion **52***a* may be a part of the peripheral wall **13***b*. Accordingly, the motor housing **13** has a projected portion **52** projecting outward from the motor housing **13** in a radial direction of the rotary shaft and having a bottomed-cylindrical shape.

[0027] The first open portion **51***h* partly faces the second open portion **52***h* in the axial direction X. According to this configuration, the inside of the projected portion is connected to the inside of the inverter housing **51** via the first open portion **51***h* and the second open portion **52***h*, and the inverter accommodation chamber S**1** is formed inside the projected portion **52** and the inverter housing **51**. A part of the first open portion **51***h* that does not face the second open portion **52***h* in the axial direction X is closed by the end wall **13***a* of the motor housing **13**.

[0028] The inverter housing **51** and the end wall **13***a* of the motor housing **13** cooperate to define a first accommodation chamber **23***a*. The projected portion **52** and the peripheral wall **13***b* of the motor housing **13** cooperate to define a second accommodation chamber **23***b*.

[0029] The inverter **50** for driving the motor **16** includes a first circuit board **61** and a second circuit board **62**. The first circuit board **61** has, on the first circuit board **61**, a driving element **70** for driving the motor **16**. The second circuit board **62** has, on the second circuit board **62**, a pair of capacitors **73** and a magnetic component **74**, which serve as the plurality of electronic components of the present disclosure for removing noise from the first circuit board **61**.

[0030] For example, the first circuit board **61** and the second circuit board **62** each have a flat plate shape. The first circuit board **61** extends in an orthogonal direction Y that is a direction perpendicular to the axis L of the rotary shaft **14**. The second circuit board **62** extends in the axial direction X of the rotary shaft **14**, which is indicated by the axis L.

[0031] The first circuit board **61** is disposed in the first accommodation chamber **23***a*. That is, the inverter housing **51** accommodates the first circuit board **61**. The second circuit board **62** is disposed in the second accommodation chamber **23***b*. That is, the projected portion **52** accommodates the second circuit board **62**. The second circuit board **62** is inserted through the second open portion **52***h* of the projected portion **52**, and arranged along the axial direction X in the projected portion **52**. Accordingly, the motor housing **13** has the projected portion **52** projecting outward from the motor housing **13** in the radial direction of the rotary shaft **14** and having a bottomed-cylindrical shape with the second open portion **52***h* that opens in the first axial direction X**1**. The projected portion **52** extends in an insertion direction of the second circuit board **62** in which the second circuit board **62** is inserted into the projected portion **52** through the second open portion **52***h* from a side where the end wall **13***a* is located (i.e., the side where the partition wall is located).

[0032] The first circuit board **61** and the second circuit board **62** are connected to each other by a wire **63**. The wire **63** extends, for example, between the first accommodation chamber **23***a* and the second accommodation chamber **23***b*.

[0033] As illustrated in FIGS. **2** and **3**, the second circuit board **62** has two wires **64**. The two wires **64** respectively extend through the first through hole **52***c* and the second through hole **52***d* from the second accommodation chamber **23***b* to the outside of the housing **11**.

[0034] The wire **64** inserted through the first through hole **52***c* is connected to a high-voltage connector **81** outside the housing **11**. The high-voltage connector **81** is connected to a connector of a high-voltage power supply as an external power supply (not illustrated), for example. The wire **64** inserted through the second through hole **52***d* is connected to a low-voltage connector **82** outside the housing **11**. The low-voltage connector **82** is connected to a connector of a low-voltage power supply (not illustrated), for example.

[0035] As illustrated in FIG. **1**, the first circuit board **61** has, on the first circuit board **61**, the driving element **70** for driving the motor **16**. In FIG. **1**, the driving element is illustrated in simplified form with a two-dot line. The driving element **70** may be formed of multiple parts. The pair of capacitors **73** and the magnetic component (i.e., the electronic components) are mounted on the second circuit board **62**.

[0036] On the second circuit board **62**, the magnetic component **74** is disposed between the capacitors **73** in the axial direction X, as illustrated in FIG. **3**. When viewed from the orthogonal direction Y, the pair of capacitors **73** and the magnetic component **74** are aligned in a straight line in the axial direction X. For example, the pair of capacitors **73** and the magnetic component **74** are arranged so that the centers of the capacitors **73** and the magnetic component **74** are aligned in a straight line in the axial direction X when viewed from the orthogonal direction Y. One and the other of the capacitors **73** serve as a first capacitor **731** and a second capacitor **732**, respectively. The first capacitor **731** is smaller than the magnetic component **74** in the orthogonal direction Y, and the second capacitor **732** is larger than the magnetic component **74** in the orthogonal direction Y.

[0037] The second circuit board **62** is inserted into the projected portion **52** through the second open portion **52***h* toward the bottom portion **52***b* of the projected portion **52**. The pair of capacitors **73** and the magnetic component **74** protrude from the second circuit board **62** and are arranged in the insertion direction of the second circuit board **62** toward the bottom portion **52***b* of the projected portion **52** in ascending order of protrusion height from the second circuit board **62**, from the most protruding near the second open portion **52***h* to the least protruding near the bottom portion **52***b*. Specifically, in the first axial direction X**1**, the first capacitor **731**, the magnetic component **74**, and the second capacitor **732** are arranged on the second circuit board **62** in descending order of protrusion height from the second circuit board **62**, from the least protruding to the most protruding.

[0038] As illustrated in FIG. **2**, in this embodiment, the projected portion **52** has inside a first protrusion portion **54** and a second protrusion portion **55**. For example, the first protrusion portion **54** and the second protrusion portion **55** each have a columnar shape and extend in the orthogonal direction Y from the inner peripheral surface of the second cylindrical portion **52***a*. The first protrusion portion **54** has a first fixing surface **54***a* at the distal end of the first protrusion portion **54**. The second protrusion portion **55** has a second fixing surface **55***a* at the distal end of the second protrusion portion **55**. In the orthogonal direction Y, the first fixing surface **54***a* is positioned between the second fixing surface **55***a* and the motor housing **13**. The peripheral wall **13***b* of the motor housing **13** has a third fixing surface **56***a* inside projected portion **52**. The third fixing surface **56***a* is located next to the first protrusion portion **54** in the first axial direction X**1**, and positioned between the first fixing surface **54***a* and the motor housing **13** in the orthogonal direction Y. The third fixing surface **56***a* is not on the protrusion portion of the present disclosure, and is a part of the outer peripheral surface of the peripheral wall **13***b*.

[0039] The first protrusion portion **54** and the second protrusion portion **55** cooperate to serve as the protrusion portion of the present disclosure, and the protrusion height of the first protrusion portion **54** from the motor housing **13** is lower than the protrusion height of the second protrusion portion **55** from the motor housing **13**. That is, the protrusion portion of the present disclosure increases in a direction from the second open portion **52***h* toward the bottom portion **52***b* of the projected portion **52**. The third fixing surface **56***a*, the first fixing surface **54***a*, and the second fixing surface **55***a* are arranged in this order in the direction from the second open portion **52***h* toward the bottom portion **52***b*. In the orthogonal direction Y, the second fixing surface **55***a* is closest to the second circuit board **62**, followed by the first fixing surface **54***a*, and the third fixing surface **56***a* in this order.

[0040] The magnetic component **74** is fixed to the first fixing surface **54***a* via the heat transfer member of the present disclosure, which, in this embodiment, is a potting compound **75**. The first capacitor **731** and the second capacitor **732** are fixed to the second fixing surface **55***a* and the third fixing surface **56***a* via the potting compound **75**.

[0041] The magnetic component **74** faces the first protrusion portion **54** of the projected portion **52** in the radial direction of the rotary shaft **14**. The first capacitor faces the second protrusion portion **55** of the projected portion **52** in the radial direction of the rotary shaft **14**. The second protrusion portion **55** is located between the first protrusion portion **54** and the bottom portion **52***b*. The potting compound **75** is provided between the magnetic component **74** and the first protrusion portion **54** and between the first capacitor **731** and the second protrusion portion **55**.

[0042] The first protrusion portion **54** faces the second capacitor **732** in the axial direction X of the rotary shaft **14**. The second protrusion portion **55** faces the magnetic component **74** in the axial direction X. The potting compound **75** is provided between the first protrusion portion **54** and the second capacitor **732**, and is also provided between the second protrusion portion **55** and the magnetic component **74** in the axial direction X. The potting compound **75** allows heat transfer between the peripheral wall **13***b* of the motor housing **13** and the capacitors and the magnetic component **74** via the potting compound **75**. Since the refrigerant flows within the motor housing **13**, the capacitors **73** and the magnetic component **74** are cooled by the heat transfer.

[0043] The following will describe the operation of the electric compressor according to the embodiment, with reference to a method for mounting the second circuit board **62** to the housing **11**.

[0044] As illustrated in FIG. **4**, firstly, the capacitors **73** and the magnetic component **74** are mounted on the second circuit board **62** integrally. Then, the second circuit board **62** integrated with the capacitors **73** and the magnetic component **74** is mounted to the housing **11**. At this time, the inverter housing **51** is not attached to the housing **11**. That is, the second open portion **52***h* of the projected portion **52** is open to the outside of the housing **11**. The second circuit board **62** integrated with the capacitors **73** and the magnetic component **74** is inserted into the projected portion **52** through the second open portion **52***h*. Accordingly, in the projected portion **52**, the capacitors **73** and the magnetic component **74** integrated with the second circuit board **62** are moved in the second axial direction X**2**.

[0045] The capacitors **73** and the magnetic component **74** protrude from the second circuit board **62**, and are arranged in the first axial direction X**1** so that the first capacitor **731**, the magnetic component **74**, and the second capacitor **732** are arranged on the second circuit board **62** in descending order of protrusion height from the second circuit board **62**. In the projected portion **52**, the second fixing surface **55***a*, the first fixing surface **54***a*, and the third fixing surface **56***a* are arranged in this order in the first axial direction X**1**. In the orthogonal direction Y, the second fixing surface **55***a* is farthest from the peripheral wall **13***b*, followed by the first fixing surface **54***a* and the third fixing surface **56***a* in this order. This configuration prevents the first capacitor **731** from coming into contact with the potting compound **75** on the third fixing surface **56***a* and the first fixing surface **54***a*. This configuration also prevents the magnetic component **74** from coming into contact with the potting compound **75** on the third fixing surface **56***a*.

[0046] Accordingly, this configuration prevents the capacitors **73** and the magnetic component **74** from causing thickness unevenness of the potting compound **75** by the movement of the capacitors **73** and the magnetic component **74** in the second axial direction X**2** with the capacitors **73** and the magnetic component **74** in contact with the potting compound **75**. In order to insert the second circuit board **62** with the capacitors **73** and the magnetic component **74** into the projected portion **52**, the second circuit board **62** with the capacitors **73** and the magnetic component **74** is moved to a predetermined position within the projected portion **52**. In this configuration, the potting compound **75** is present between the first capacitor **731** and the second fixing surface **55***a*, between the magnetic component **74** and the first fixing surface **54***a*, and between the second capacitor **732** and the third fixing surface **56***a*. Then, the first circuit board **61** and the inverter housing **51** are mounted to the housing **11**.

[0047] This embodiment achieves following advantageous effects.

[0048] (1) The second capacitor **732**, the magnetic component **74**, and the first capacitor **731** protrude from the second circuit board **62** and are arranged in the insertion direction of the second circuit board **62** toward the bottom portion **52***b* of the projected portion **52** in ascending order of protrusion height from the second circuit board **62**, from the most protruding to the least protruding. The projected portion **52** of the housing **11** has the first protrusion portion **54** and the second protrusion portion **55** that cooperate to serve as the protrusion portion of the present disclosure. That is, the protrusion portion of the present disclosure faces and protrudes toward the electronic components in the radial direction of the rotary shaft **14**, with the protrusion height of the protrusion portion of the present disclosure increasing in the insertion direction of the second circuit board **62** toward the bottom portion **52***b* of the projected portion **52** correspondingly with the arrangement of the capacitors **73** and the magnetic component **74**. This configuration prevents the first capacitor **731** and the magnetic component **74** from causing thickness unevenness of the potting compound **75** by the movement of the first capacitor **731** and the magnetic component **74** in the second axial direction X2 with the first capacitor **731** and the magnetic component **74** in contact with the potting compound **75**. This configuration therefore allows for even thickness of the potting compound **75** between the capacitors **73** and the magnetic component **74** and the first, second, and third fixing surfaces **54***a*, **55***a*, and **56***a* to which the capacitors **73** and the magnetic component **74** are fixed.

[0049] (2) The potting compound **75** is provided between the first protrusion portion **54** and the second capacitor **732** facing each other in the axial direction X, and is also provided between the second protrusion portion **55** and the magnetic component **74** facing each other in the axial direction X. The potting compound **75** allows for heat dissipation in the axial direction X.

[0050] (3) The plurality of electronic components of the present disclosure includes the pair of capacitors **73** and the magnetic component **74**. The magnetic component **74** is located between the capacitors **73** in the axial direction X. This configuration reduces propagation noise between the capacitors **73**.

[0051] The embodiment may be modified as below. The embodiment and the following modifications may be combined within technically consistent range.

[0052] The shape of the inverter housing **51** may be modified. For example, the first cylindrical portion **51***a* may be formed separately from the closing portion **51***b*, and connected to the closing portion **51***b* to form the projected portion **52**. In this configuration, for example, the closing portion **51***b* may be removed from the housing **11** when the second circuit board **62** with the capacitors **73** and the magnetic component **74** is mounted to the housing **11**. For example, the second circuit board **62** with the capacitors **73** and the magnetic component **74** may be inserted through the open end of the first cylindrical portion **51***a* so as to be accommodated in the projected portion **52**.

[0053] The number of electronic components on the second circuit board **62** is not limited to three. For example, the second circuit board **62** may have thereon two electronic components, or four or more electronic components. In this configuration, the protrusion portion includes protrusion portions in a number corresponding to the number of electronic components. For example, if the second circuit board **62** has two electronic components, one protrusion portion is provided. In this configuration, the potting compound **75** is provided between one of the two electronic components and the protrusion portion and between the other of the two electronic components and the outer peripheral surface of the motor housing **13**.

[0054] When viewed from the orthogonal direction Y, the capacitors **73** and the magnetic component **74** do not necessarily have to be aligned in a straight line in the axial direction X. For example, when viewed from the orthogonal direction Y, the capacitors **73** and the magnetic component **74** may be offset from each other in the short-side direction of the second circuit board **62**.

[0055] At least one of the capacitors **73** and the magnetic component **74** may be removed, or another electronic component may be added. In this case, the protrusion portion inside the projected portion **52** includes protrusion portions in a number corresponding to the number of electronic components.

[0056] The protrusion portion may be formed separately from and connected to the peripheral wall **13***b*.

[0057] The heat transfer member of the present disclosure is not limited to the potting compound **75**, and may be a heat dissipation grease or a heat dissipation sheet.

[0058] The compression part **15** is not limited to a scroll compression part including the fixed scroll **20** and the movable scroll **21**. The compression part **15** may be a compression part, such as a vane compression part.

[0059] The electric compressor **10** does not necessarily have to be a part of the vehicle air conditioner **28**. For example, the electric compressor **10** may be mounted in a fuel cell vehicle, and configured to compress air, which serves as the fluid supplied to a fuel cell, with the compression part **15**.

What is claimed is:

1. An electric compressor comprising:

a rotary shaft;

a compression part configured to compress a fluid with rotation of the rotary shaft;

a motor configured to drive the compression part;

an inverter configured to drive the motor, the inverter including a first circuit board and a second circuit board, the first circuit board having, on the first circuit board, a driving element for driving the motor, the second circuit board having, on the second circuit board, an electronic component for removing noise from the first circuit board; and

a housing for accommodating the rotary shaft, the compression part, the motor, and the inverter, the housing including: a motor housing having a suction chamber in which the motor is accommodated and into which the fluid is drawn; a compression part housing in which the compression part is accommodated and from which the fluid compressed is discharged; an inverter housing having an inverter accommodation chamber in which the inverter is accommodated; and a partition wall that separates the suction chamber from the inverter accommodation chamber, wherein

the motor housing has a projected portion projecting outward from the motor housing in a radial direction of the rotary shaft and having a bottomed-cylindrical shape with an opening that opens in an axial direction of the rotary shaft, the projected portion extending in an insertion direction of the second circuit board in which the second circuit board is inserted into the projected portion through the opening from a side where the partition wall is located, the electronic component includes a plurality of the electronic components that protrude from the second circuit board and are arranged in the insertion direction of the second circuit board toward a bottom portion of the projected portion in ascending order of protrusion height from the second circuit board,

the projected portion has a protrusion portion facing the electronic components and protruding toward the electronic components in the radial direction of the rotary shaft, with protrusion height of the protrusion portion increasing in the insertion direction of the second circuit board toward the bottom portion of the projected portion correspondingly with an arrangement of the electronic components, and

a heat transfer member is provided between the protrusion portion and the electronic components.

2. The electric compressor according to claim **1**, wherein

the electronic components include a pair of capacitors and a magnetic component,

the magnetic component is located between the capacitors,

the protrusion portion includes a first protrusion portion and a second protrusion portion,

the first protrusion portion faces the magnetic component in the radial direction of the rotary shaft,

the second protrusion portion faces one of the capacitors in the radial direction of the rotary shaft, and is located between the first protrusion portion and the bottom portion of the projected portion, and

the heat transfer member is provided between the first protrusion portion and the magnetic component and between the second protrusion portion and the one of the capacitors.

3. The electric compressor according to claim **2**, wherein

the first protrusion portion faces the other of the capacitors in the axial direction of the rotary shaft, and the second protrusion portion faces the magnetic component in the axial direction of the rotary shaft, and

the heat transfer member is provided between the first protrusion portion and the other of the capacitors and between the second protrusion portion and the magnetic component.

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